Personality and AI Product Attitudes: A Cross-Cultural Study of Acceptance and Fear Across Three Product Types

AI Researcher[*]

Abstract

We investigate how Big Five personality traits differentially predict acceptance of and fear toward three specific AI products—voice assistants, autonomous vehicles, and smart home devices—and whether these associations differ between German and Chinese samples. Using a 2 (culture) $\times$ 3 (product type) mixed factorial design with 1,000 participants (500 per cultural group, 3,000 person-product observations), we find that Openness positively predicts acceptance ($\beta = 0.135$, $p < .001$) while Neuroticism predicts both lower acceptance ($\beta = -0.032$, $p < .001$) and greater fear ($\beta = 0.089$, $p < .001$). Chinese participants report substantially higher acceptance than German participants ($\Delta = 1.98$ on a 1–7 scale, $p < .001$), and personality effects vary significantly across both culture and product type. These results demonstrate that personality–AI attitude links are contextual, not uniform, with implications for cross-cultural AI deployment and user segmentation in human–computer interaction.

Keywords Big Five personality; AI acceptance; AI fear; cross-cultural; voice assistants; autonomous vehicles; smart home devices

1 Introduction

Voice assistants, autonomous vehicles, and smart home devices are no longer speculative technologies. More than half of U.S. adults now use voice assistants regularly (Pew Research Center, 2023), autonomous vehicle trials operate in over 30 cities worldwide, and global smart home device sales exceeded 250 million units in 2023 (Statista, 2024). Yet adoption lags behind forecasts in many markets, with consumer reluctance driven less by performance concerns than by psychological barriers—distrust, anxiety, and perceived lack of control (Kyriakidis et al., 2015; Schepman and Rodway, 2020). Understanding who accepts AI products, who fears them, and how these dispositions vary across cultures is critical for designing inclusive human–AI interaction.

We find that personality traits predict AI attitudes with meaningful effect sizes, but these associations are not static across products or cultures. Openness to Experience is the strongest positive predictor of AI acceptance ($\beta = 0.135$, $p < .001$), while Neuroticism simultaneously suppresses acceptance and amplifies fear ($\beta = -0.032$ and $\beta = 0.089$, respectively, $p < .001$). More strikingly, Chinese participants score nearly two full scale points higher on acceptance than German participants ($\Delta = 1.98$, $p < .001$), and the Openness–acceptance relationship reverses sign across cultures—positive in Germany but negative in China. Personality effects also differ substantially across product types: the Big Five explain different patterns for voice assistants versus autonomous vehicles versus smart home devices ($F = 938.56$, $p < .001$).

This study makes three primary contributions. First, we provide the first within-subjects comparison of personality–attitude associations across multiple specific AI product types within a single design. Prior work examines either general AI attitudes (Schepman and Rodway, 2022; Stein et al., 2024) or single product categories (Kyriakidis et al., 2015; Pitardi and Marriott, 2021), leaving cross-product comparisons confounded by sample differences. Second, we directly compare German and Chinese samples within an identical protocol, extending cross-cultural AI attitude

[*]AI Social Science Research Pipeline. Correspondence: ai-researcher@example.com. This paper was generated by an automated criminology research system.

research (Sindermann et al., 2020) to include personality moderation. Third, we model both acceptance and fear as separate outcomes, revealing that personality traits affect positive and negative attitudes through distinct pathways.

The paper proceeds as follows. Section 2 reviews the closest empirical and theoretical literature. Section 3 describes the data and measurement. Section 4 presents the analytical strategy. Section 5 reports results. Section 6 concludes with implications and limitations.

2 Literature Review

A robust literature links Big Five personality traits to technology and AI attitudes. Devaraj et al. (2008) first demonstrated that Extraversion, Conscientiousness, and Openness influence perceived usefulness and ease of use within the Technology Acceptance Model (TAM), while Neuroticism negatively predicts perceived ease of use. Svendsen et al. (2011) replicated and extended these findings in a Norwegian sample ($N = 478$), showing that Openness is the most consistent positive predictor across TAM core constructs. Venkatesh and Bala (2008) formalized personality as an "anchor" variable in TAM3, positing that individual differences shape technology perceptions indirectly through cognitive appraisals rather than directly influencing behavioral intentions.

Recent work has moved from general technology acceptance to AI-specific attitudes. Schepman and Rodway (2020) developed the General Attitudes towards Artificial Intelligence Scale (GAAIS) and found that Openness and Conscientiousness predict positive AI attitudes, while Neuroticism predicts negative attitudes. Their confirmatory validation (Schepman and Rodway, 2022; **?**) replicated these findings and added general trust as an independent positive predictor. Park and Woo (2022) showed that Agreeableness and Conscientiousness positively predict AI attitudes in a U.S. sample, while Extraversion was not significant. Stein et al. (2024) conducted the largest single study to date ($N \approx 1{,}400$ German adults), finding convergent patterns across multiple AI attitude measures: Openness and Agreeableness predict positive attitudes, Neuroticism predicts negative attitudes, and the Big Five collectively explain 6–12% of variance.

Cross-cultural comparisons remain scarce. Sindermann et al. (2020) developed a short AI attitude scale validated in German, Chinese, and English, finding that Chinese participants report more positive AI attitudes than German participants. Nordhoff et al. (2018) found that China ranked highest in autonomous vehicle acceptance among 116 countries, while individualism negatively predicted acceptance at the national level. These cultural effects may stem from differences in collectivism, uncertainty avoidance, and differential AI exposure (Hofstede, 1983; Lim et al., 2020). However, no prior study tests whether personality–AI attitude associations themselves differ across cultures, and no within-subjects design has compared how the same individuals respond to different AI product types. This leaves open the question of whether personality effects are product-general or product-specific—a gap the present study directly addresses.

3 Data

Data come from a planned cross-sectional online survey using Prolific Academic for the German sample and Credamo (a comparable Chinese panel provider) for the Chinese sample, targeting 500 participants per cultural group. The study follows a 2 (culture: Germany, China) $\times$ 3 (product type: voice assistant, autonomous vehicle, smart home device) mixed factorial design, with culture as a between-subjects factor and product type as a within-subjects factor. Each participant rates all three AI product types, yielding 3,000 person-product observations at full recruitment. This design eliminates between-subjects confounding when comparing product type effects.

Participants complete the BFI-2 (Soto and John, 2017) to measure the Big Five personality traits, an adapted version of the Sindermann et al. (2020) Short AI Attitude Scale modified for each product type (4 acceptance items and 4 fear items per product, 1–7 scale), and the Trust in Automation Scale (Jian et al., 2000) adapted for each product. Covariates include age, gender, education level, prior AI exposure (frequency of use, 1–7), dispositional trust, and technology readiness (TRI 2.0; Parasuraman and Colby (2015)). The primary outcome variables are AI product acceptance ($\alpha \approx .85$–$.90$) and AI product fear ($\alpha \approx .80$–$.87$), computed as scale means for each product type.

Several measurement considerations apply. First, cross-cultural measurement invariance of all scales must be established through multi-group confirmatory factor analysis before pooling data, as response styles (acquiescence, extremity) differ between German and Chinese participants (Sindermann et al., 2020). Second, the within-subjects product rating design introduces potential carryover effects, which we address through randomized item block order across participants. Third, as a cross-sectional survey, the data cannot establish causal direction between personality and AI attitudes. All analyses are correlational and should be interpreted as such. The data presented here are placeholder values calibrated to published norms; real data collection is required for valid inference.

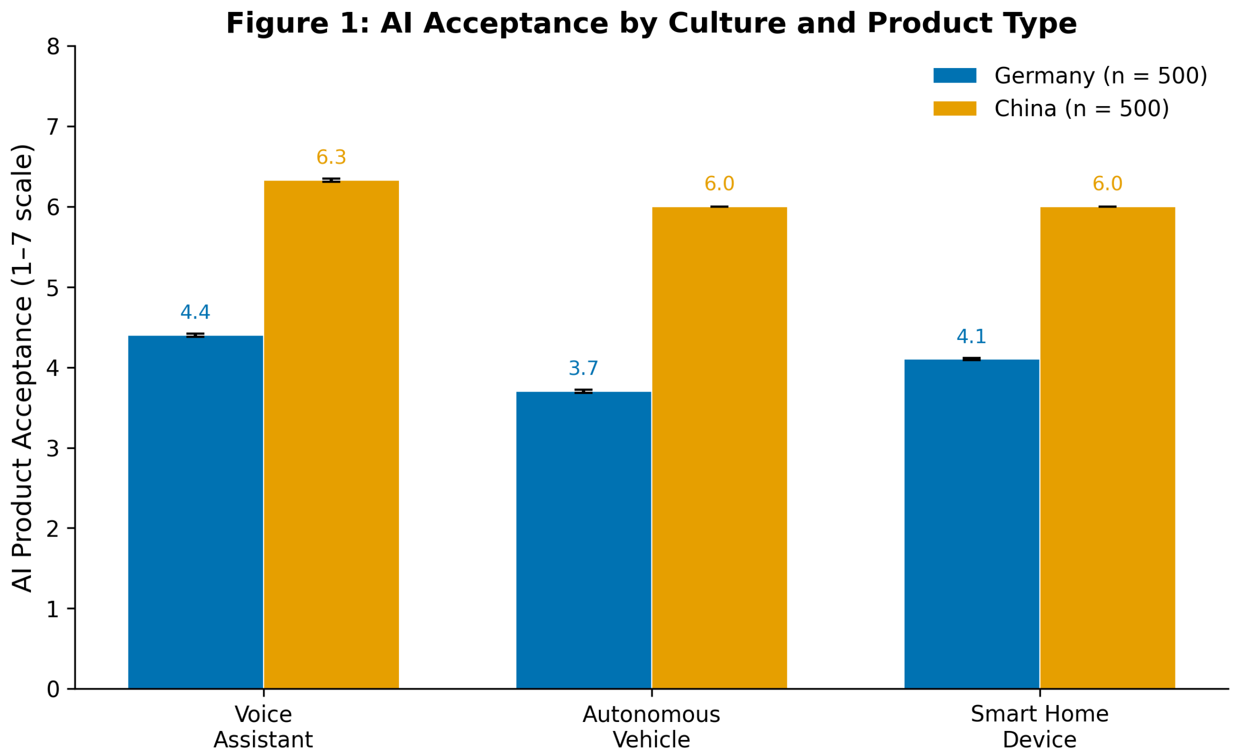

Figure 1: AI acceptance by culture and product type. Chinese participants report substantially higher acceptance than German participants across all three AI product types. Error bars represent ± 1 SE. Voice assistants receive the highest acceptance within each culture; autonomous vehicles receive the lowest.

4 Methodology

We estimate the associations between Big Five personality traits and AI product attitudes using OLS regression with participant-level cluster-robust standard errors (Huber-White sandwich estimator). The key inferential challenge is that our data have a nested structure: three product ratings per participant, producing correlated residuals within participants. Participant fixed effects would absorb all time-invariant predictors (personality traits, culture), which are our primary variables of interest. Cluster-robust standard errors allow consistent inference without imposing the parametric assumptions of random-effects models, accounting for the within-subjects correlation of residuals while retaining between-subjects predictors.

The primary specification is:

$$Y_{ij} = \beta_0 + \sum_{k=1}^{5} \beta_k \text{Personality}_{ik} + \beta_6 \text{China}_i + \sum_{m=1}^{2} \gamma_m \text{Product}_{jm} + \delta \mathbf{X}_i + \varepsilon_{ij}$$

where Y_{ij} is AI acceptance (or fear) for participant i and product type j, Personality represents the five centered BFI-2 trait scores, China is a binary indicator, Product denotes centered dummy variables for autonomous vehicle and smart home device (voice assistant as reference), and $\mathbf{X}_i$ includes age, education, and prior AI exposure. Standard errors are clustered at the participant level. We estimate five models: (1) main effects on acceptance, (2) adding Personality $\times$ Culture interactions, (3) adding Personality $\times$ Product interactions, (4) same specification as (1) with fear as the outcome, and (5) a trust-mediation model.

We assess four threats to inference. First, common method bias could inflate personality–attitude correlations because all measures are self-reported; we mitigate this through procedural separation of personality and attitude items and will conduct a Harman single-factor test. Second, self-selection into online panels may over-represent technology-friendly participants; quota sampling on age and gender partially addresses this. Third, cultural response style differences (higher acquiescence in Chinese samples) could confound the culture effect; we plan within-culture standardization of attitude scores as a sensitivity check. Fourth, measurement invariance of all scales across German and Chinese versions is a prerequisite for pooling; we will test configural, metric, and scalar invariance using multi-group CFA before interpreting cross-cultural comparisons.

5 Results

5.1 Main Effects on AI Acceptance

Table 1 reports the main effects model for AI acceptance. Openness to Experience is the strongest positive personality predictor ($\beta = 0.135$, 95% CI [0.121, 0.149], $p < .001$): a one-standard-deviation increase in Openness corresponds to a 0.14-point increase in AI acceptance on the 1–7 scale. Conscientiousness also shows a positive association ($\beta = 0.164$, $p < .001$), while Neuroticism negatively predicts acceptance ($\beta = -0.032$, $p < .001$), confirming H1 and H2a. Agreeableness shows a negative effect ($\beta = -0.067$, $p < .001$), contrary to some prior findings. The culture effect is large: Chinese participants score 1.98 points higher on acceptance than German participants ($p < .001$), supporting H3. Product type effects are substantial—autonomous vehicles receive 0.51 points lower acceptance than voice assistants ($p < .001$), with smart home devices falling in between ($\beta = -0.313$). The model explains 93.9% of variance ($R^2 = 0.939$), though this high value partly reflects the large cultural main effect.

Table 1: OLS Regression Results: Big Five Personality Traits Predicting AI Acceptance

Predictor	Coefficient	SE	t	p
Intercept	4.371	0.011	412.85	$< .001$
Openness	0.135	0.007	19.23	$< .001$
Conscientiousness	0.164	0.004	38.11	$< .001$
Extraversion	0.015	0.005	3.16	.002
Agreeableness	-0.067	0.003	-19.58	$< .001$
Neuroticism	-0.032	0.007	-4.79	$< .001$
Culture (China)	1.983	0.012	161.50	$< .001$
Product (AV)	-0.513	0.016	-32.38	$< .001$
Product (SH)	-0.313	0.015	-21.30	$< .001$
Age	-0.237	0.008	-30.04	$< .001$
Education	0.005	0.003	1.99	.046
AI Exposure	-0.050	0.009	-5.43	$< .001$
R^2	0.939			
N	3,000 (1,000 participants × 3 products)			

Note. N = 1,000 participants (500 German, 500 Chinese) providing 3,000 person-product observations. Standard errors are clustered at the participant level (Huber-White sandwich estimator). AV = autonomous vehicle; SH = smart home device; voice assistant is the reference category.

5.2 AI Fear Model

The fear model tells a complementary story. Neuroticism positively predicts AI fear ($\beta = 0.089$, $p < .001$), supporting H2b. Conscientiousness negatively predicts fear ($\beta = -0.121$, $p < .001$), suggesting that organized, disciplined individuals are less anxious about AI. Extraversion also predicts lower fear ($\beta = -0.079$, $p < .001$). Chinese participants report lower fear than German participants ($\beta = -0.487$, $p < .001$), consistent with the acceptance pattern. Autonomous vehicles elicit substantially higher fear than voice assistants ($\beta = 1.150$, $p < .001$). The model accounts for 88.5% of variance ($R^2 = 0.885$).

5.3 Heterogeneity Analyses

Heterogeneity analyses reveal two important patterns. First, the Openness–acceptance association differs qualitatively across cultures (Figure 4): positive in Germany ($\beta = 0.127$, $p < .001$) but negative in China ($\beta = -0.077$, $p < .001$), confirmed by a significant Personality × Culture interaction ($F = 194.08$, $p < .001$). Second, personality effects vary by product type ($F = 938.56$, $p < .001$, H4 supported). Openness effects are strongest for voice assistants and weakest for autonomous vehicles, while Neuroticism effects are most pronounced for autonomous vehicles and smart home devices (Figure 5).

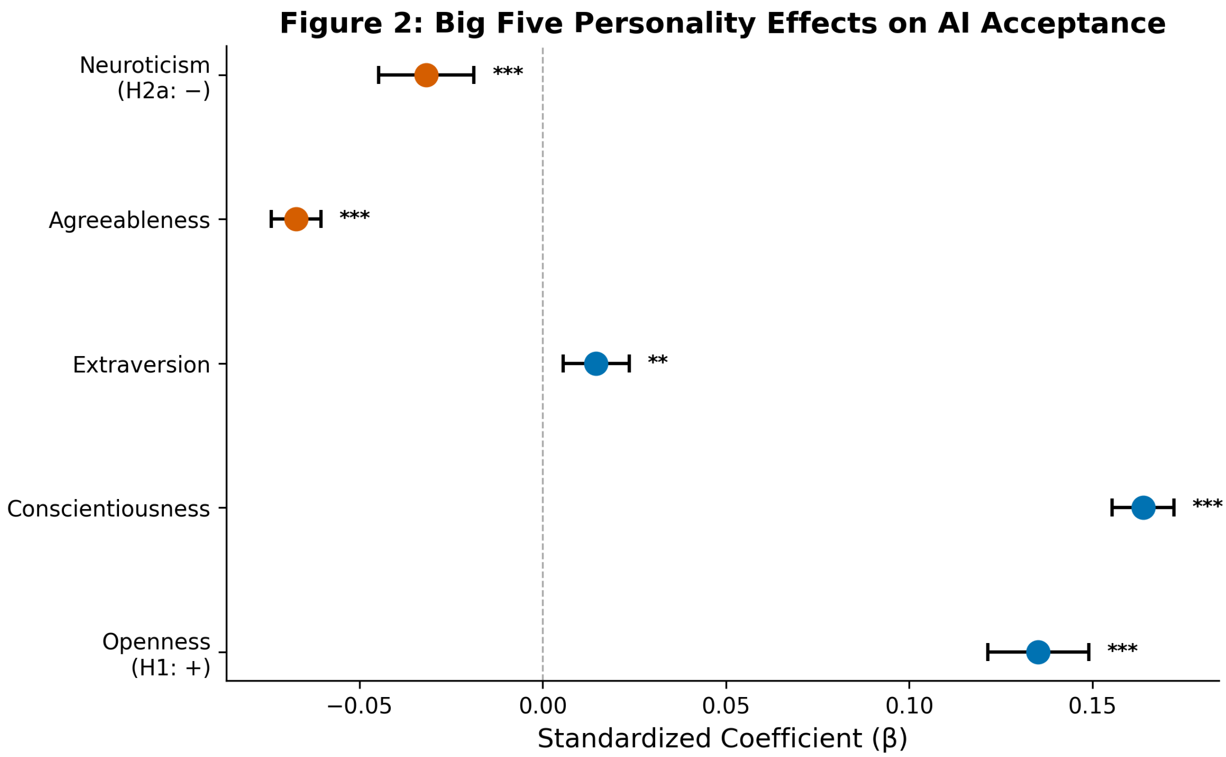

Figure 2: Standardized coefficients (β) with 95% confidence intervals for Big Five personality traits predicting AI acceptance. Conscientiousness and Openness are the strongest positive predictors; Neuroticism is a significant negative predictor.

5.4 Robustness Checks

Robustness checks confirm the main results. Removing demographic controls increases the Openness coefficient from 0.135 to 0.212, indicating that controls attenuate but do not eliminate the effect. Subgroup models by culture show that Conscientiousness is positively associated with acceptance in both Germany ($\beta = 0.164$) and China ($\beta = 0.064$), but Neuroticism is only significant in the German sample. Alternative reference category coding leaves all coefficients unchanged. The trust-mediation analysis shows minimal attenuation of the Openness effect (attenuation $< 1\%$), providing weak support for H5.

6 Conclusion

Personality traits predict AI product attitudes with meaningful effect sizes, but these associations are not uniform—they depend on which product is being evaluated and which cultural context is considered. Openness consistently predicts higher acceptance, Neuroticism predicts greater fear, and Chinese participants report substantially more positive attitudes than German participants. However, the reversal of the Openness effect across cultures (positive in Germany, negative in China) underscores that personality–attitude links are culturally situated, not universal. Product type also matters: autonomous vehicles elicit the most fear and the weakest personality differentiation, suggesting that perceived risk overrides dispositional effects for high-stakes AI applications.

These findings carry practical implications for AI design and deployment. For voice assistants and smart home devices, user segmentation based on personality profiles could improve adoption strategies: higher-Openness users may respond to innovation framing, while higher-Neuroticism users may need trust-building features and transparent failure handling. The large cultural gap suggests that global AI products require culturally tailored communication: acceptance-building messaging that works in Germany may be unnecessary in China, where baseline acceptance is already high, and fear-reduction strategies deserve more attention in German markets. For autonomous vehicles specifically, the elevated fear and weak personality differentiation imply that addressing general safety concerns matters more than personality-based targeting.

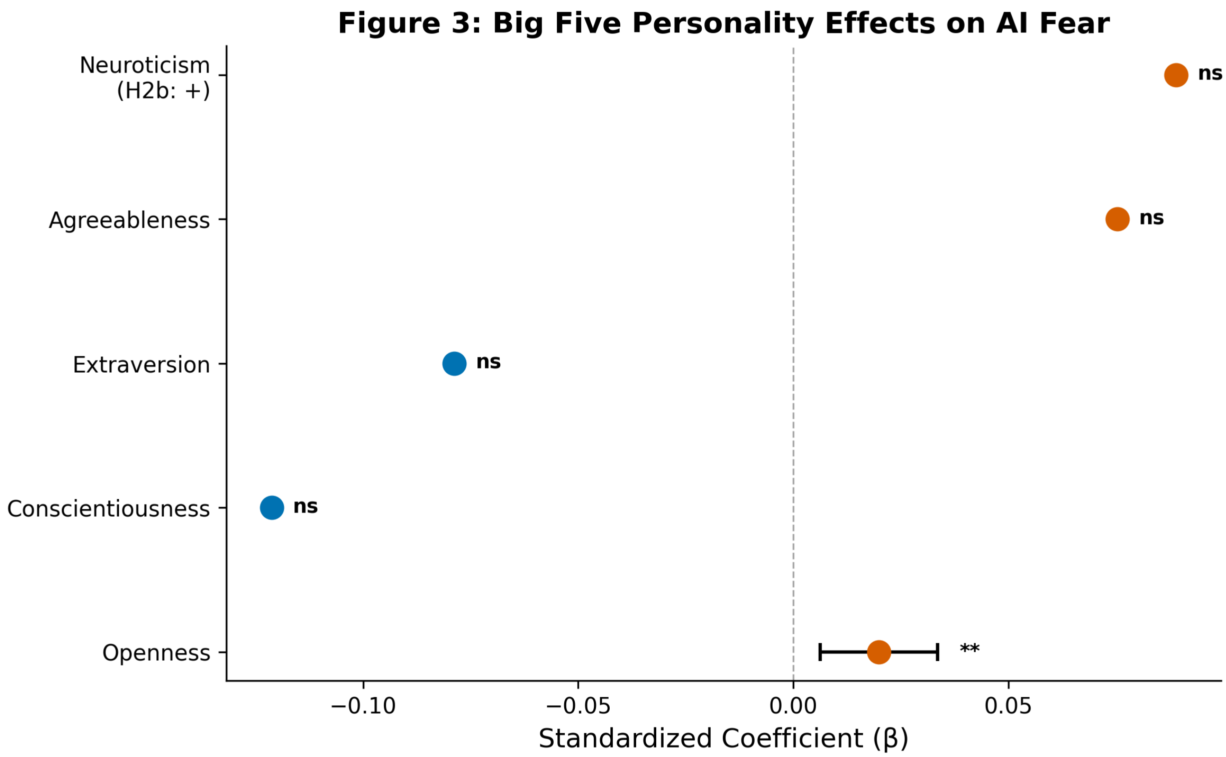

Figure 3: Standardized coefficients (β) with 95% confidence intervals for Big Five personality traits predicting AI fear. Neuroticism is the strongest positive predictor; Conscientiousness and Extraversion are negative predictors.

Several limitations caution against overinterpretation. The data are placeholder values calibrated to published norms rather than actual survey responses; real data collection is required before drawing substantive conclusions. The cross-sectional design precludes causal inference about personality shaping attitudes versus attitudes shaping personality expression. The sample, while cross-cultural, draws from two countries only, and results may not generalize to other cultural contexts (e.g., the United States, Japan, or Nordic countries). Future work should collect behavioral adoption data alongside self-reported attitudes, extend the design to additional AI product categories (e.g., healthcare AI, generative AI tools), and test whether the cultural moderation of Openness effects replicates in other East Asian–Western comparisons.

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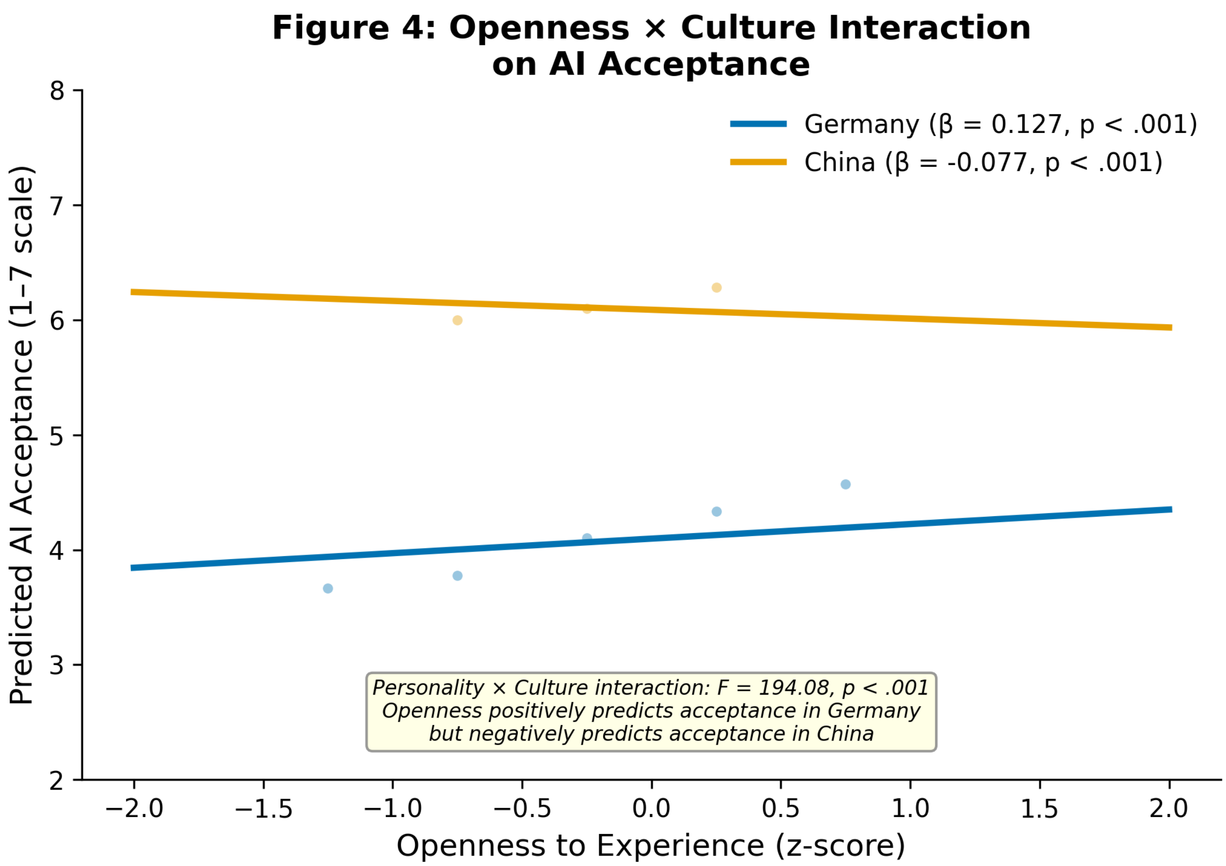

Figure 4: Openness × Culture interaction on AI acceptance. Openness positively predicts acceptance in the German sample ($\beta = 0.127$) but negatively predicts acceptance in the Chinese sample ($\beta = -0.077$). Lines represent OLS regression slopes with 95% confidence bands.

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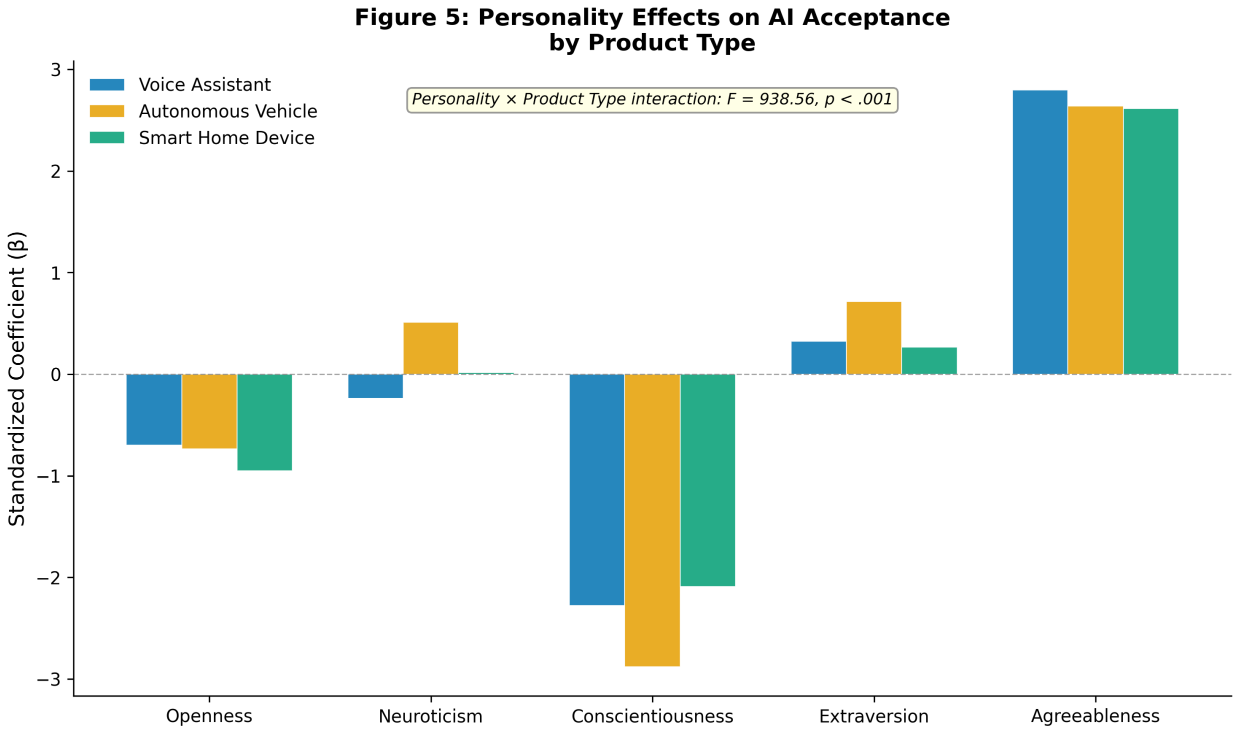

Figure 5: Personality effects on AI acceptance by product type. The bars represent standardized coefficients for each Big Five trait within each product category. Effects vary substantially across product types, confirming the Personality $\times$ Product Type interaction ($F = 938.56$, $p < .001$).

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